

二、解答题 (7-12 题, 每题 7 分, 共 42 分)

7. 设 $\begin{cases} x = \ln \sin t, \\ y = \sin t - t \cos t, \end{cases}$, 求 $\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}}$ 和 $\left. \frac{d^2y}{dx^2} \right|_{t=\frac{\pi}{3}}$.

$$\left. \frac{dy}{dx} \right|_{t=\frac{\pi}{3}} = \left. \frac{t \sin^2 t}{\cos t} \right|_{t=\frac{\pi}{3}} = \frac{\pi}{2}, \quad (2 \text{ 分})$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{(\sin^2 t + 2t \sin t \cos t) \cos t + t \sin^3 t}{\cos^2 t} \cdot \frac{1}{\frac{\sin t}{\cos t}} \\ &= \frac{\sin^2 t \cos t (\sin t + 2t \cos t) + t \sin^4 t}{\cos^3 t}, \end{aligned} \quad (5 \text{ 分})$$

从而

$$\left. \frac{d^2y}{dx^2} \right|_{t=\frac{\pi}{3}} = \frac{\frac{3}{4} \left(\frac{\sqrt{3}}{2} + 2 \cdot \frac{\pi}{3} \cdot \frac{1}{2} \right) \frac{1}{2} + \frac{\pi}{3} \left(\frac{\sqrt{3}}{2} \right)^4}{\left(\frac{1}{2} \right)^3} = \frac{3\sqrt{3}}{2} + \frac{5\pi}{2}. \quad (7 \text{ 分})$$

8. 求解一阶微分方程 $x \frac{dy}{dx} = y(\ln y - \ln x)$.

解: 整理, 原方程为齐次方程

$$\frac{dy}{dx} = \frac{y}{x} \ln \frac{y}{x}, \quad (2 \text{ 分})$$

令 $u = \frac{y}{x}$, 方程化为

$$x \frac{du}{dx} = u(\ln u - 1), \quad (4 \text{ 分})$$

分离变量, 两边积分, 得

$$\int \frac{1}{u(\ln u - 1)} du = \int \frac{1}{x} dx, \quad (5 \text{ 分})$$

解得 $u = e^{1+cx}$, 即 $y = xe^{1+cx}$. (7 分)

9. 求解二阶微分方程 $y'' + y = x$.

解: (1) 齐次微分方程 $y'' + y = 0$ 的通解为

$$y = c_1 \cos x + c_2 \sin x; \quad (3 \text{ 分})$$

(2) 求原方程本身的一个特解 $y^* = ax + b$,

代入原方程, 由待定系数法, 得 $y^* = x$; (6 分)

(3) 原方程的通解为

$$y = c_1 \cos x + c_2 \sin x + x. \quad (7 \text{ 分})$$

10. 求曲线 $y = 1 - x^2$ 在第一象限与坐标轴所围图形绕 x 轴旋转一周所得立体体积.

$$\begin{aligned} V &= \int_0^1 \pi (1 - x^2)^2 dx = \pi \int_0^1 (1 - 2x^2 + x^4) dx \\ \text{解: } &= \pi \left(1 - \frac{2}{3} + \frac{1}{5} \right) = \frac{8}{15} \pi. \end{aligned} \quad (7 \text{ 分})$$

11. 求函数 $y = 2x^3 - 3x^2$, $-1 \leq x \leq 4$ 的极值.

解: $y' = 6x^2 - 6x$, $-1 \leq x \leq 4$,

令 $y' = 0$ 得 $x_1 = 0$, $x_2 = 1$, (4 分)

又 $y'' = 12x - 6$, $-1 \leq x \leq 4$,

由 $y''(0) = -6$ 知 $y(0) = 0$ 为极大值;

$y''(1) = 6$ 知 $y(1) = -1$ 为极小值. (7 分)

12. 求曲线 $y = \int_0^x \sqrt{\sin 2t} dt$ ($0 < x < \frac{\pi}{4}$) 的弧长.

解:

$$s = \int_0^{\frac{\pi}{4}} \sqrt{1 + y'^2(x)} dx = \int_0^{\frac{\pi}{4}} \sqrt{1 + \sin 2x} dx \quad (3 \text{ 分})$$

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} (\sin x + \cos x) dx = (\sin x - \cos x) \Big|_0^{\frac{\pi}{4}} \\ &= 1. \end{aligned} \quad (7 \text{ 分})$$